

EEAIC assignment1

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1 Task

Following design strategy described in tutorial 1, we found the width that allowed for the g_m to be equal to 10 mS at the given inversion conditions.

g_m/I_D	25 (Weak Inversion)	15 (Moderate Inversion)	7.5 (Strong Inversion)
g_m [mS]	10	10	10
I_D/W [A/m]	0.064	11.36	21.45
I_D [mA]	0.4	1	1.33
W [μm]	6237	88	62

Table 1: *nch hvt* operating regions comparison

2 Task

Design Strategy: The OTA design is based on the optimization of the Unity Gain Frequency (f_u) and Phase Margin (PM). We assume that C_c establishes the dominant pole, which contributes a 90° phase shift at f_u . To meet the requirement of $PM > 64^\circ$, the following condition for the second pole (f_{p2}) must be satisfied:

$$90^\circ - \arctan\left(\frac{f_u}{f_{p2}}\right) > 64^\circ \implies f_{p2} > \frac{f_u}{0.487} \quad (1)$$

Given a target $f_u = 1.5$ MHz, we assume $f_{p2} \approx 4$ MHz. Using the approximation $f_{p2} \approx \frac{g_{m2}}{2\pi C_L}$, we derive $g_{m2} = 125 \mu\text{S}$. To maintain operation in weak inversion, the transconductance efficiency is set to $(g_m/I_D)_2 = 20 V^{-1}$, yielding a second-stage bias current $I_{D2} = 6.25 \mu\text{A}$.

Following the methodology from *Systematic Design of Analog CMOS Circuits*, a rule of thumb for the compensation capacitor C_c is about 20% of C_L , resulting in $C_c = 1$ pF. The transconductance of the input stage is then calculated as:

$$g_{m1} = 2\pi \cdot f_u \cdot C_c = 9.42 \mu\text{S} \quad (2)$$

Setting $(g_m/I_D)_1 = 20 \text{ V}^{-1}$ for the input pair $M_{1a,b}$ to stay in weak inversion, the tail current I_{M5} is approximately 950 nA. The total current consumption is $I_{D2} + I_{M5} \approx 7.2 \text{ }\mu\text{A}$, comfortably meeting the power specification of $P < 20 \text{ }\mu\text{W}$.

To minimize the systematic input offset, the transistors are sized such that $V_{DS3a} = V_{DS3b}$, ensuring that the gates of M_2 , M_{3a} , and M_{3b} see the same potentials. To do this, since the current density of M_2 is fixed by the weak inversion condition, we sized the widths of M_{3a} and M_{3b} to carry our desired current with the same current density $\frac{I_d}{W}$ ratio of M_2 , thus ensuring the same V_{OD} .

To minimize parasitic gate resistance, M_2 was implemented using a multi-finger layout; specifically, a 4-finger configuration was employed for this transistor to reduce the effective gate resistance.

Furthermore, the W/L ratios of the current source transistors M_4 and M_5 are designed to carry the desired current from the ideal current mirror relation:

$$\frac{I_{ref}}{(\frac{W}{L})_{ref}} = \frac{I}{(\frac{W}{L})} \quad (3)$$

Simulating the circuit with the $\frac{W}{L}$, we noticed that the mirror transistors carried less current than they should. Thus we enlarged a bit the width of both M_5 and M_4 to make them match our desired I_d .

The primary design objective for M_4 and M_5 was to minimize voltage headroom by maximizing their respective aspect ratios (W/L). To further enhance voltage gain, it would be ideal to increase the channel length (L) of M_4 to improve its output resistance r_o and mitigate Channel Length Modulation. In the final schematic, M_4 and M_5 maintain the same channel length to simplify layout. This represents a design trade-off where the priority for uniform sizing and low headroom was balanced against the potential for higher intrinsic gain through a longer M_4 channel.

The compensation resistor is initially set to $R_c = 1/g_{m2} = 8 \text{ k}\Omega$ to eliminate the Right-Half-Plane (RHP) zero. Post-simulation analysis (top-right corner of Figure 2) confirms that g_{m1} is highly consistent with analytical calculations. However, g_{m2} was extracted at $101 \text{ }\mu\text{S}$, showing a slight deviation attributed to the limited output resistance r_o of M_2 and especially M_4 as a consequence of the short channel length utilized. To preserve the zero-cancellation condition ($R_c \approx 1/g_{m2}$), the compensation resistance should be refined to $R_c \approx 9.9 \text{ k}\Omega$.

The following figures illustrate the complete circuit schematic and the simulation testbench, including the final device dimensions.

From the Bode analysis, the amplifier achieves a unity-gain frequency (f_u) of 1.254 MHz with a corresponding phase margin (PM) of 68.4° . As shown in the scalar output summary (top-right corner of Fig. 2), all auxiliary design constraints are satisfied. These include the transconductance efficiency (g_m/I_D) targets for the input differential pair and M_2 , the achieved transconductances, and the total power consumption, which remains within the specified $20 \text{ }\mu\text{W}$.

budget. The design successfully meets all performance requirements outlined in the assignment.

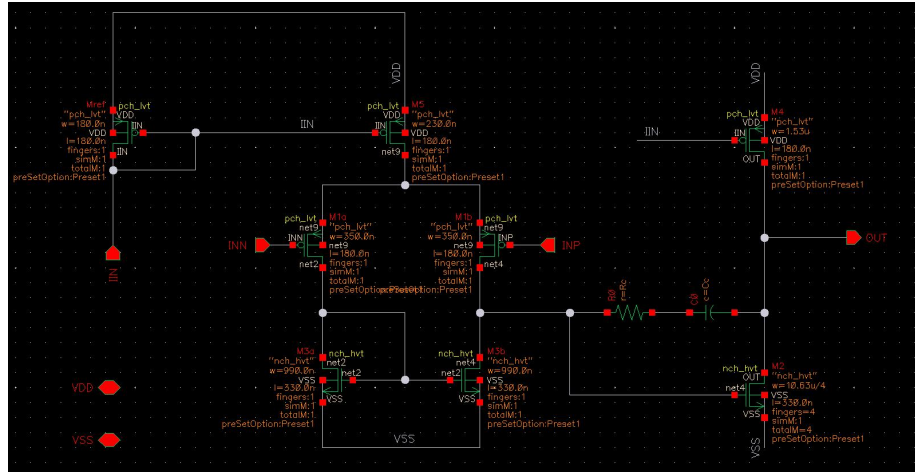


Figure 1: Schematic with relative sizings

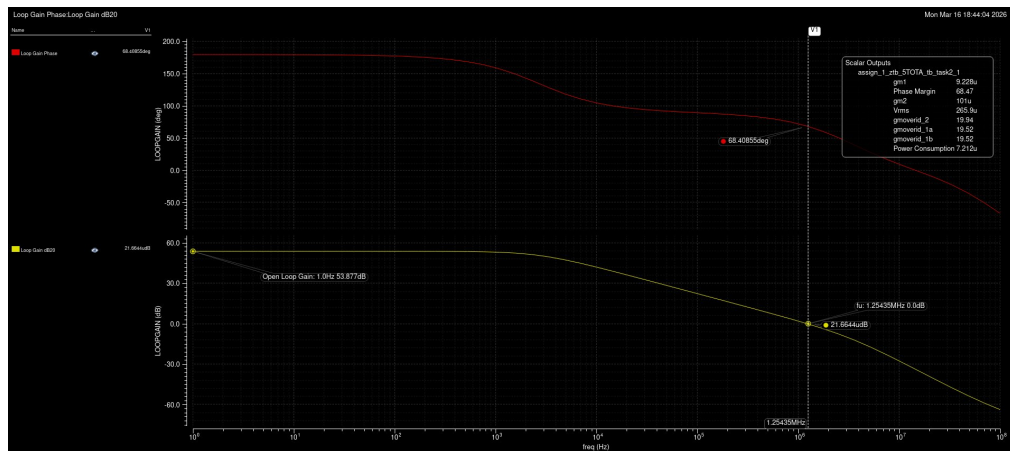


Figure 2: Bode plot with scalar outputs

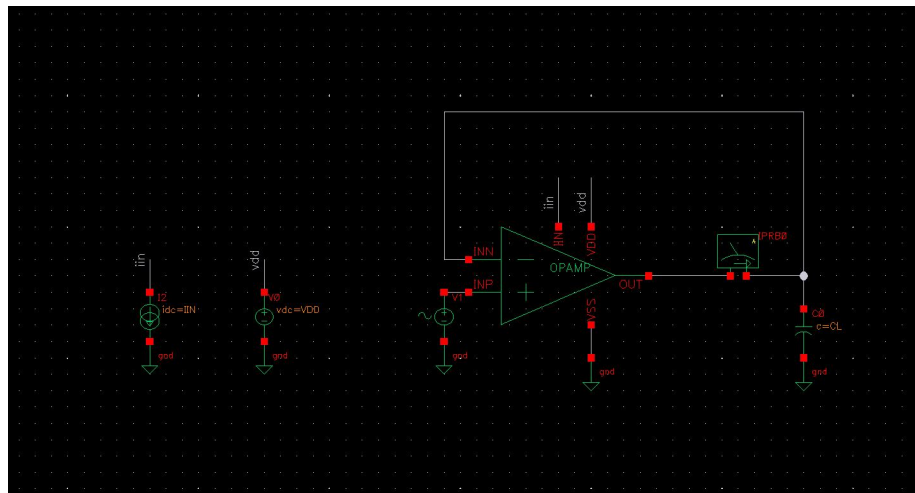


Figure 3: Testbench used

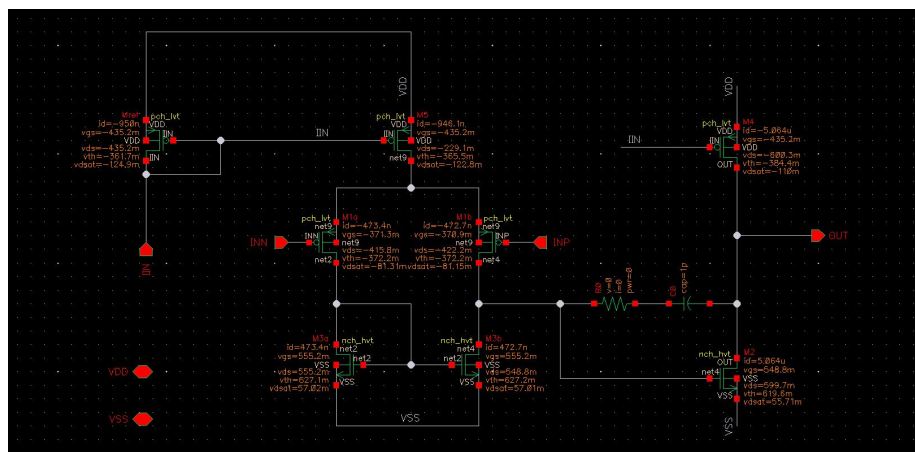


Figure 4: DC operating points

3 Task

The compensation resistor R_c , also known as the *nulling resistor*, plays a crucial role in managing the Right-Half-Plane (RHP) zero inherently created through C_c . An RHP zero is particularly problematic because it decreases the phase by 90° , significantly reducing the phase margin and the stable bandwidth.

The location of the zero s_z is governed by the following relationship:

- **Without Nulling Resistor ($R_c = 0$):** The zero is fixed in the RHP:

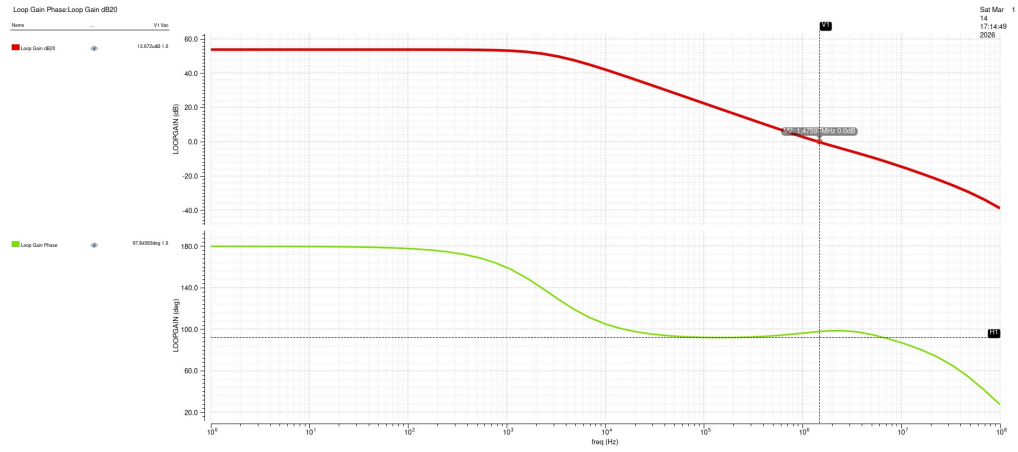
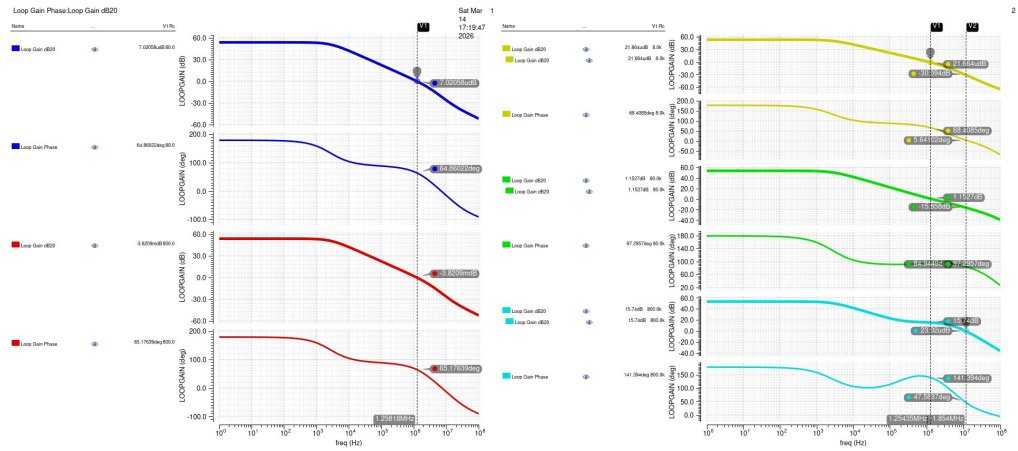
$$s_z = \frac{g_{m2}}{C_c} \quad (4)$$

- **With Nulling Resistor ($R_c > 0$):** The zero location is shifted according to:

$$s_z = \frac{-1}{C_c \left(R_c - \frac{1}{g_{m2}} \right)} \quad (5)$$

By properly selecting R_c , the zero can be moved to the Left-Half-Plane (LHP) to enhance stability, or placed at infinity by matching $R_c = 1/g_{m2}$.

To investigate these dependencies, a parametric sweep of R_c was conducted using the values $80\ \Omega$, $800\ \Omega$, $8\ \text{k}\Omega$, $80\ \text{k}\Omega$, and $800\ \text{k}\Omega$. This analysis evaluates how the zero migrates across the s -plane and its subsequent impact on the magnitude and phase response in the Bode plots.

Figure 5: Bode plot considering $R_c = 80k\Omega$ Figure 6: Bode plot sweeping R_c value = (80, 800, 8k, 80k, 800k) Ω

4 Task

An evaluation of the initial design revealed that the input-referred noise (IRN) density significantly exceeded the $70 \text{ nV}/\sqrt{\text{Hz}}$ at 1 kHz limit specified in Task 4.

Consequently, a second design iteration was performed with a primary focus on satisfying this stringent noise constraint.

In this revised design, the sizing strategy prioritized the reduction of flicker and thermal noise components. To achieve this, the gate areas of the critical transistors—specifically the input pair (M_1) and the active load (M_3)—were substantially increased. While certain dimensions were intentionally oversized to provide a conservative margin of safety and ensure the noise floor remained comfortably below the $70 \text{ nV}/\sqrt{\text{Hz}}$ threshold, we recognize that further iterations could be conducted to optimize the trade-off between silicon area and noise performance.

It is important to note that this improvement in noise performance did not come at the expense of other key specifications. The optimized amplifier maintains a high open-loop gain and exhibits robust stability, as confirmed by the Phase Margin. The resulting circuit is a stable amplifying stage with a higher unity-gain frequency (f_u) of 3.77 MHz. A particularly noteworthy result is the relationship between bandwidth and noise density in this design. Despite the unity-gain frequency being approximately three times higher than that of the first design (increasing from 1.25 MHz to 3.77 MHz), which theoretically integrates more total noise power, the input-referred noise spectral density is significantly lower. This demonstrates that the strategic increase in transconductance and device area successfully outweighed the noise contributions typically associated with increased bandwidth.

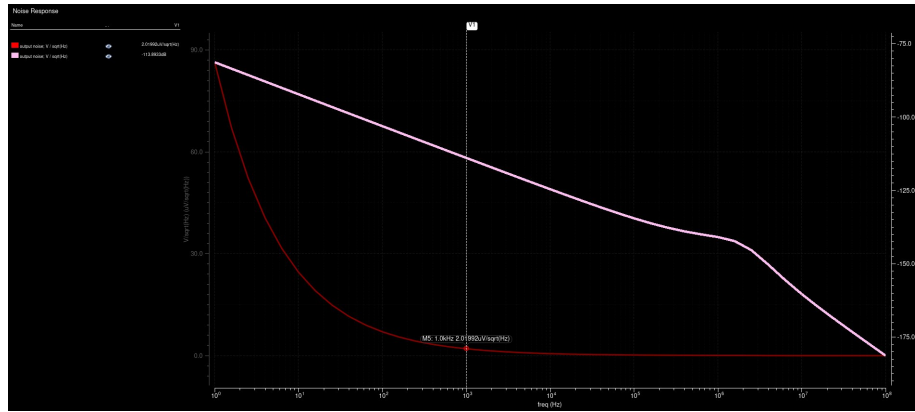


Figure 7: Noise in the original design (Task 2) both in linear and dB scale

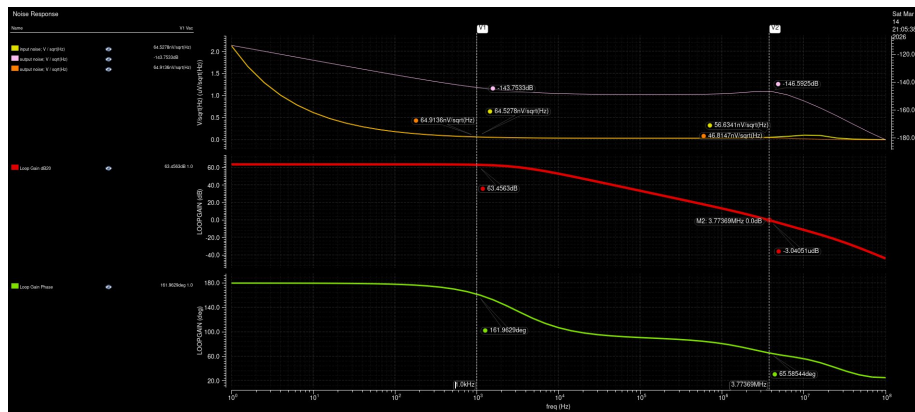


Figure 8: Noise in the new design in linear and dB scale

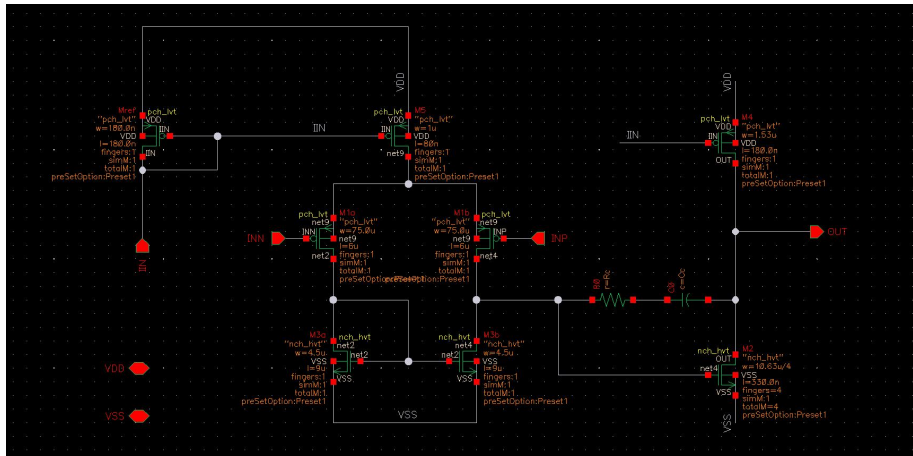


Figure 9: Schematic of the new design

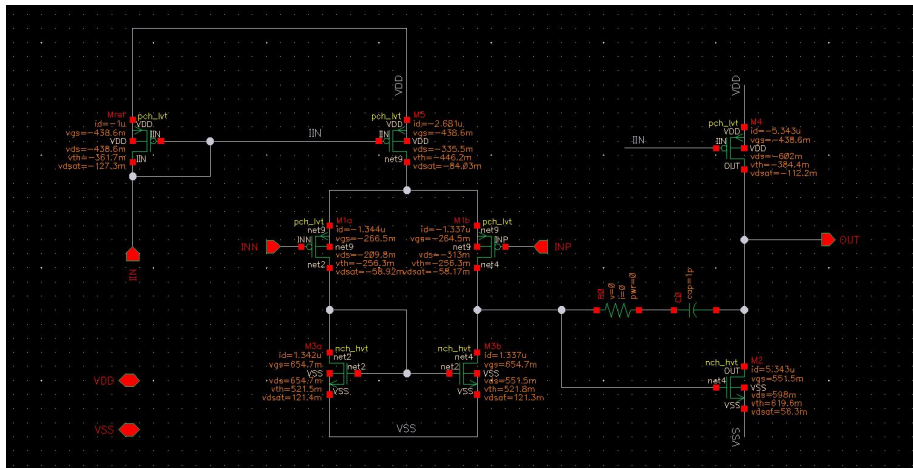


Figure 10: DC operating points of the new design

5 Task

From the definition of Noise Efficiency Factor:

$$NEF := V_{rms,in} \sqrt{\frac{2 \cdot I_{tot}}{\pi \cdot V_T \cdot 4kT \cdot f_{BW}}}$$

To calculate the NEF, the integrated noise was obtained using the Virtuoso Calculator by integrating the input-referred noise power over the amplifier's bandwidth. The total current was extracted from the DC operating points of the tail current source (M_5) and the second-stage branch (M_4).

- Initial Design:* In the initial design, the noise performance was secondary to meeting the basic gain and power specifications.
 Integrated Noise ($V_{rms,in}$): $265.9 \mu\text{V}$ ($V_{rms} = \text{sqrt}(\text{integ}((\text{getData}("in"?result "noise"))**2) 1 1.25\text{M}))$).
 Current Consumption: $I_{M5} = 946.1 \text{ nA}$, $I_{M4} = 5.064 \mu\text{A} \implies I_{tot} \approx 6.01 \mu\text{A}$.
 Calculated NEF: 22.48
- Design from Task 4:* The design was specifically optimized to meet the $70 \text{ nV}/\sqrt{\text{Hz}}$ requirement.
 Integrated Noise ($V_{rms,in}$): $86.51 \mu\text{V}$ ($V_{rms} = \text{sqrt}(\text{integ}((\text{getData}("in"?result "noise"))**2) 1 3.77\text{M}))$).
 Current Consumption: $I_{M5} = 2.681 \mu\text{A}$, $I_{M4} = 5.343 \mu\text{A} \implies I_{tot} \approx 8.02 \mu\text{A}$.
 Calculated NEF: 4.87

Note that to ensure the accuracy of these results, a sanity check was performed by comparing the values obtained via the Virtuoso Calculator integration with the *Noise Summary* tool. The results showed great agreement, confirming that the calculated $V_{rms,in}$ and subsequent NEF values are reliable indicators of the circuit's improved power-noise efficiency.

While the initial design met all gain, bandwidth, and power specifications, its Noise Efficiency Factor of 22.48 is impractical for real-world sensor interfaces. This high value indicates a poor trade-off between power consumption and noise performance. In contrast, the second design iteration successfully optimized the power-noise balance, reducing the NEF to 4.87 and delivering a significantly more viable solution for energy-efficient applications.

Energy-efficient Analog Integrated Circuits

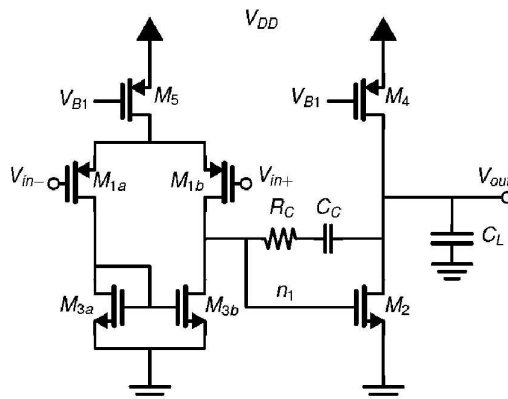
Sensor Interface – Exercise 1

Assignment

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1. Consider the design strategy described in Section 6 of the tutorial. Repeat the analysis for an n-channel device `nch_hvt`. Assume $L = 330$ nm and a drain-source voltage $V_{DS} = 0.6$ V. Design the input pair to achieve $g_m = 10$ mS in the case of $g_m/I_D = 25$, $g_m/I_D = 10$, and $g_m/I_D = 7.5$. Hint: you can now sweep the gate voltage V_G and fix the source voltage V_S .
2. The schematic of a two-stage operational transconductance amplifier is shown below. Assume the amplifier operates with a supply $V_{DD} = 1.2$ V and the input DC level is $V_{DD}/2$. Design the amplifier to meet the following specifications and explain your design strategy carefully:
 - Unity-gain frequency $f_u > 1$ MHz
 - Open-loop gain $A_{OL} > 45$ dB
 - Load capacitance $C_L = 5$ pF
 - Phase Margin $PM > 64^\circ$
 - Power consumption < 20 μ W
 - Bias M_{1a} , M_{1b} and M_2 to operate in weak inversion: $g_{m1}/I_{D1} = g_{m2}/I_{D2} = 20$
 - Minimize the systematic input offset
 - Assume $R_C = 1/g_{m2}$



3. What is the effect of the compensation resistance R_C on the circuit transfer function? Can you improve the phase margin by using a different value of R_C ?
4. Run a noise analysis and simulate the input-referred noise. Modify the design targeting an input-referred noise lower than $70 \text{ nV}/\sqrt{\text{Hz}}$ at 1 kHz (you don't need to satisfy the specifications from 2).
5. Simulate the noise efficiency factor of the two-stage amplifier.